



SIMATS
ENGINEERING



SIMATS
Saveetha Institute of Medical And Technical Sciences
(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Lab experiment-15

Decision Tree using Python

Aim

To write a Python program to implement a Decision Tree classification algorithm.

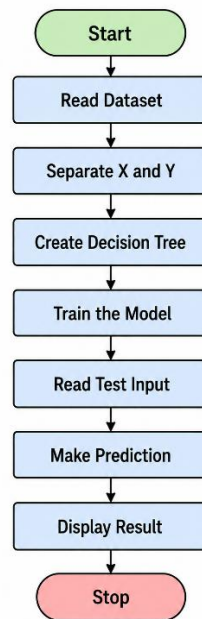
Objective

To build a Decision Tree model.

Algorithm

1. Start.
2. Read the dataset.
3. Separate input features and target values.
4. Create a Decision Tree classifier.
5. Train the model using the dataset.
6. Read test input from the user.
7. Predict the class.
8. Display the result.
9. Stop.

Flowchart



Code

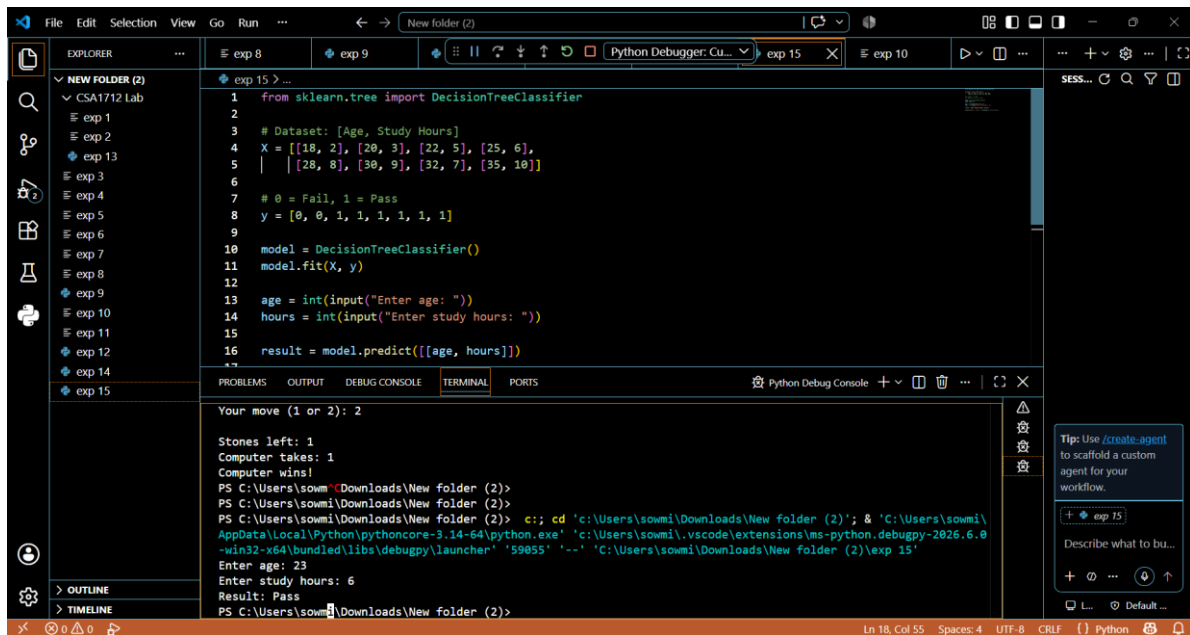
```
from sklearn.tree import DecisionTreeClassifier

# Dataset: [Age, Study Hours]
X = [[18, 2], [20, 3], [22, 5], [25, 6],
      [28, 8], [30, 9], [32, 7], [35, 10]]
# 0 = Fail, 1 = Pass
y = [0, 0, 1, 1, 1, 1, 1, 1]

model = DecisionTreeClassifier()
model.fit(X, y)

age = int(input("Enter age: "))
hours = int(input("Enter study hours: "))
result = model.predict([[age, hours]])
print("Result:", "Pass" if result[0] == 1 else "Fail")
```

Output



The screenshot shows a VS Code editor with a Python script in the main editor and its execution output in the terminal. The script defines a dataset, trains a Decision Tree classifier, and takes user input to predict a result. The terminal shows the execution of the script, including the input of age 23 and study hours 6, resulting in a 'Pass' prediction.

```
1 from sklearn.tree import DecisionTreeClassifier
2
3 # Dataset: [Age, Study Hours]
4 X = [[18, 2], [20, 3], [22, 5], [25, 6],
5      | [28, 8], [30, 9], [32, 7], [35, 18]]
6
7 # 0 = Fail, 1 = Pass
8 y = [0, 0, 1, 1, 1, 1, 1, 1]
9
10 model = DecisionTreeClassifier()
11 model.fit(X, y)
12
13 age = int(input("Enter age: "))
14 hours = int(input("Enter study hours: "))
15
16 result = model.predict([[age, hours]])
```

Terminal Output:

```
Your move (1 or 2): 2
Stones left: 1
Computer takes: 1
Computer wins!
PS C:\Users\sowmi\Downloads\New folder (2)>
PS C:\Users\sowmi\Downloads\New folder (2)>
PS C:\Users\sowmi\Downloads\New folder (2)> c:; cd 'c:\Users\sowmi\Downloads\New folder (2)'; & 'C:\Users\sowmi\AppData\Local\Python\pythoncore-3.14-64\python.exe' 'c:\Users\sowmi\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\launcher' '59055' '-.' 'C:\Users\sowmi\Downloads\New folder (2)\exp 15'
Enter age: 23
Enter study hours: 6
Result: Pass
PS C:\Users\sowmi\Downloads\New folder (2)>
```

Result

The Decision Tree classifier was implemented successfully.

Conclusion

The program successfully classifies the input data using a Decision Tree.